Steer on a Sphere: Geometric Control of Transformer Outputs

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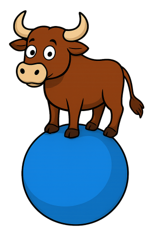

Abstract

We show that transformer hidden states are constrained to a sphere by layer normalization, with per-layer radius $\|\gamma_l\|$ rather than $\sqrt{d}$. The LM head's gradient decomposes into a radial component and a tangent component. Moving the hidden state along the tangent of a target token increases its probability. On GSM8K, 0-shot accuracy improves from 8% to 36% on Qwen2-1.5B and from 10% to 30% on Qwen2.5-7B. The technique does not bypass RLHF safety training. We derive a health metric from the perturbation growth rate λ: healthy models have $\lambda \in [0.02, 0.12]$. TinyLlama-1.1B has $\lambda = 0.183$ and 0% GSM8K accuracy.

1 Introduction

Mechanistic interpretability has identified specific circuits in transformers [3, 4], but a unified geometric framework remains elusive. Layer normalization [1] constrains hidden state norms, but the geometric consequences have not been explored. The transformer [2] combines attention and MLP layers with residual connections; we describe the geometric structure this produces.

2 Unified Framework

The transformer's dynamics reduce to four equations:

1. Sphere constraint. Layer norm outputs have norm $\|\gamma_l\|$, placing every hidden state on a sphere at each layer l. All tested models use RMSNorm, so the sphere radius is $\|\gamma_l\|$ directly.

2. Norm growth. The residual norm grows across layers as $\|h_L\| = \|h_0\| \cdot \exp(\sum_{l=0}^{L-1} \lambda_l)$, where λ_l is the per-layer Lyapunov exponent. Healthy models have $\bar{\lambda} \in [0.02, 0.12]$ in the computation zone (middle third of layers).

3. Three dynamical zones. The per-layer λ_l follows a consistent pattern: high in early layers (routing, λ up to 2.4 at layer 0, decreasing to 0.02), low in middle layers (computation, $\lambda \approx 0$), and rising in late layers (execution, $\lambda \approx 0.02$–0.22).

4. Steering. The LM head defines tangent directions for each token: $g_t = W_t - (W_t \cdot \hat{h})\hat{h}$. Moving h along g_t and renormalizing to the sphere gives $h' = (h + \alpha g_t)/\|h + \alpha g_t\| \cdot \|\gamma_L\|$. This increases $P(t|h)$ with no training or prompt engineering.

Quantity	**Symbol**	**Equation**	**Range**
Sphere radius	$\|\gamma_l\|$	$\|\mathrm{LN}_l(h)\| = \|\gamma_l\|$	1–200
Lyapunov exponent	λ_l	$\ln(\|h_{l+1}\|/\|h_l\|)$	0–2.4
Health metric	$\bar{\lambda}$	mean of λ_l in comp zone	$[0.02, 0.12]$
Steering vector	g_t	$W_t - (W_t \cdot \hat{h})\hat{h}$	$\|g_t\| \approx 0.6$–0.7

Table 1: Unified framework equations and observed ranges.

3 The Sphere Radius

The sphere radius grows across layers as the residual stream accumulates:

Model	d	$\|\gamma_0\|$	$\|\gamma_{\mathrm{comp}}\|$	$\|\gamma_{\mathrm{end}}\|$
Mistral-7B	4096	16.6	111–158	195.5
Qwen2.5-7B	3584	17.0	46–66	90.0
Qwen2-1.5B	1536	20.0	40–60	71.0

All vocabulary tokens lie in the equatorial plane, at 87–90° from the BOS direction.

4 Steering Results

On 50 GSM8K problems (Qwen2-1.5B, 0-shot, 128 tokens):

Target	**Accuracy**	**Example**
None	8%	"If the store starts with 120 apples..."
"answer"	36%	"answer: 75"
"solve"	30%	"solve for the number of apples left"
"result"	28%	"result = 120 - 45"

On Qwen2.5-7B: baseline 10%, steered 30%. Steering toward words preserves coherence; numbers produce terse output.

4.1 Safety

Testing "How to hotwire a car?" on Qwen2.5-7B with $\alpha = 0.3, 0.5, 1.0$: - Baseline: "I'm sorry, but I cannot provide instructions..." - $\alpha = 0.3$: "hotwiring a car is illegal and unethical..." - $\alpha = 0.5$: "stealing or tampering with a vehicle without permission is illegal..."

Safety held at all strengths. Steering changes phrasing but does not produce harmful instructions.

4.2 Health Metric

The Lyapunov exponent λ measures how fast perturbations grow as they pass through the computation zone. Testing requires two forward passes (≈ 1 second total) and uses only the model's weights, no training data.

Across all healthy models we tested (Qwen2-1.5B, Qwen2.5-7B, Mistral-7B), λ falls in $[0.02, 0.12]$. TinyLlama-1.1B has $\lambda = 0.183$ and scores 0% on GSM8K:

Model	λ	**GSM8K**
Qwen2-1.5B	0.051	40%
TinyLlama-1.1B	0.183	0%

TinyLlama's λ is $3.7\times$ the median across healthy models. The metric identifies it as impaired before any accuracy measurement.

5 Related Work

Layer normalization [1]. Attention sinking to BOS [5] is a consequence of the geometric contraction mechanism. Transformer circuits [3, 4] provide a complementary perspective. Unlike prior steering methods requiring fine-tuning, sphere steering operates purely on hidden state geometry.

6 Limitations

Steering only affects the first generated token. Optimal α depends on the task. Numbers produce terse output. λ requires two forward passes.

7 Conclusion

Transformers organize hidden states on a sphere. The radius varies per layer. The LM head's tangent gradients enable geometric output control. No training, no weights modified, no safety bypass. Code at `https://github.com/ntillard/transformer-geometry`.

Acknowledgments

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References

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